
Skills

- Python (Pandas, NumPy, Matplotlib, Seaborn)
- SQL · PostgreSQL · Jupyter Notebook
- Microsoft Excel · Power BI · Tableau
- Exploratory Data Analysis (EDA)
- Data Cleaning / Wrangling · Data Validation
- Descriptive Statistics · Correlation Analysis
- Time-Series Analysis · Pivot Tables
- KPI Analysis · Trend Analysis · Dashboards
- Predictive Modeling · Classification Metrics
- Git / GitHub · Stakeholder Reporting

Projects

SALES ANALYSIS — Australian Apparel Limited · Python, Pandas, Seaborn, Jupyter Q4 2020

github.com/tahaSawan/apparel-sales-analysis

- Analyzed **7,560** retail transactions (**\$340M**) with **Python/Pandas** to identify a **4.8x revenue gap** between top- and bottom-performing states across 7 regions and 4 customer segments
- Built **20+** visualizations (heatmaps, time-series trends, distributions) in **Seaborn/Matplotlib** and an executive dashboard to communicate findings to non-technical stakeholders
- Delivered **6 data-driven recommendations** on regional marketing, peak-hour staffing, and promotional timing, backed by descriptive statistics and correlation analysis

EMPLOYEE TURNOVER ANALYTICS — Predictive Retention Insights · Python, Pandas, Scikit-learn, Jupyter 2025

- Conducted **EDA** on **12K** employee records to identify satisfaction, hours, and project load as primary turnover drivers (~16.6% attrition)
- Clustered leavers into **3 risk personas** and built a **Gradient Boosting** attrition model (**AUC 0.979, 91.7% recall**)
- Converted predictions into risk zones (**Safe / Low / Medium / High**) to help HR prioritize retention for high-probability exit candidates

PREPXPRT — Education Analytics Platform · SQL, PostgreSQL, Chart.js, Dashboards 2025 – Present

- Designed a **PostgreSQL** data model capturing question attempts, accuracy, study time, mock-exam results, and course progress for NUST NET and MDCAT learners
- Built analytics dashboards to visualize accuracy trends, study activity, subject performance, and exam readiness from student performance data
- Created data-driven workflows that surface weak areas, error patterns, and revision priorities for personalized practice recommendations

Work Experience

FREELANCE DATA ANALYTICS DEVELOPER — Independent June 2025 – Present

- Analyzed learner performance datasets to surface weak subject areas, engagement trends, and progress **KPIs** using **SQL** and **Python**
- Built dashboards and reports to track data quality, learning outcomes, and usage metrics for decision support

FREELANCE DATA REPORTING SPECIALIST — Client Project, Sweden (RE:WEAVE) June 2026

- Structured market statistics, forecast figures, and stakeholder reporting sections from client requirements; delivered organized data-driven content in two days

Education

BS COMPUTER SCIENCE — FAST NUCES Islamabad · Robotics Specialization Graduate

Gold Medal for Academic Excellence